

Shannon: Every PERFECTLY SECRET SKE must have the above limitations

Thm For any SKE with PERFECT SECRECY:

$$|K| \geq |M|$$

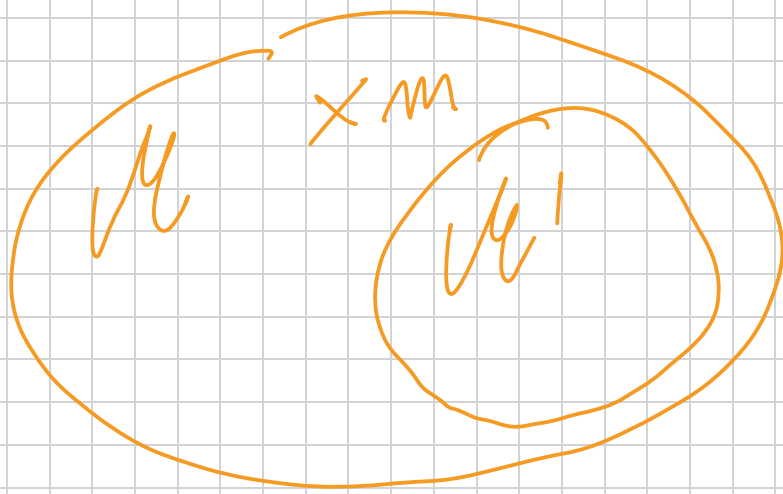
Proof: Let  $\Pi$  be ANY SKE. Fix the uniform distribution over  $M$ . Take any  $c \in \mathcal{C}$  s.t.  $\Pr[C=c] > 0$ .

Consider:  $M' = \{ \text{Dec}(k, c) : k \in K \}$

Assume that  $|K| < |M|$ .

Since  $|M'| \leq |K| < |M|$

$$\Rightarrow |M'| < |M|$$



There is  $m \in M \setminus M'$ .  
Now look:

$$\Pr [M = m] = \frac{1}{|M|}$$

On the other hand:

$$\Pr [M = m \mid C = c] = 0$$



The way: We shall do something DIFFERENT.

Our goal: Alice & Bob share 1 key of length independent of length of msg.

EXERCISE Consider an alternative def. of PERFECT  
SECRECY. We define a so-called GAME (experiment).

ADVERSARY

$A$

$$\begin{array}{l} \leftarrow c = \text{Enc}(K, m_b) \\ \hline b' \in \{0, 1\}, \end{array}$$

$\mathcal{C}$

$$K \leftarrow \mathcal{K}$$

CHALLENGER

$$\text{GAME}(b, m_0, m_1), b \in \{0, 1\}$$

$$m_0, m_1 \in \mathcal{M}.$$

We say that  $\Pi$  is PERFECTLY SECRET

if  $\forall A, \forall m_0, m_1 \in \mathcal{M} :$

$$\left| \Pr[b' = 1 : \text{GAME}_A(0, m_0, m_1)] + \right.$$

$$- \Pr [b' = 1 : \text{GAME}_A(1, m_0, m_1)] \mid$$

$$= 0$$

Alternative:  $\text{GAME}_A(0, m_0, m_1)$

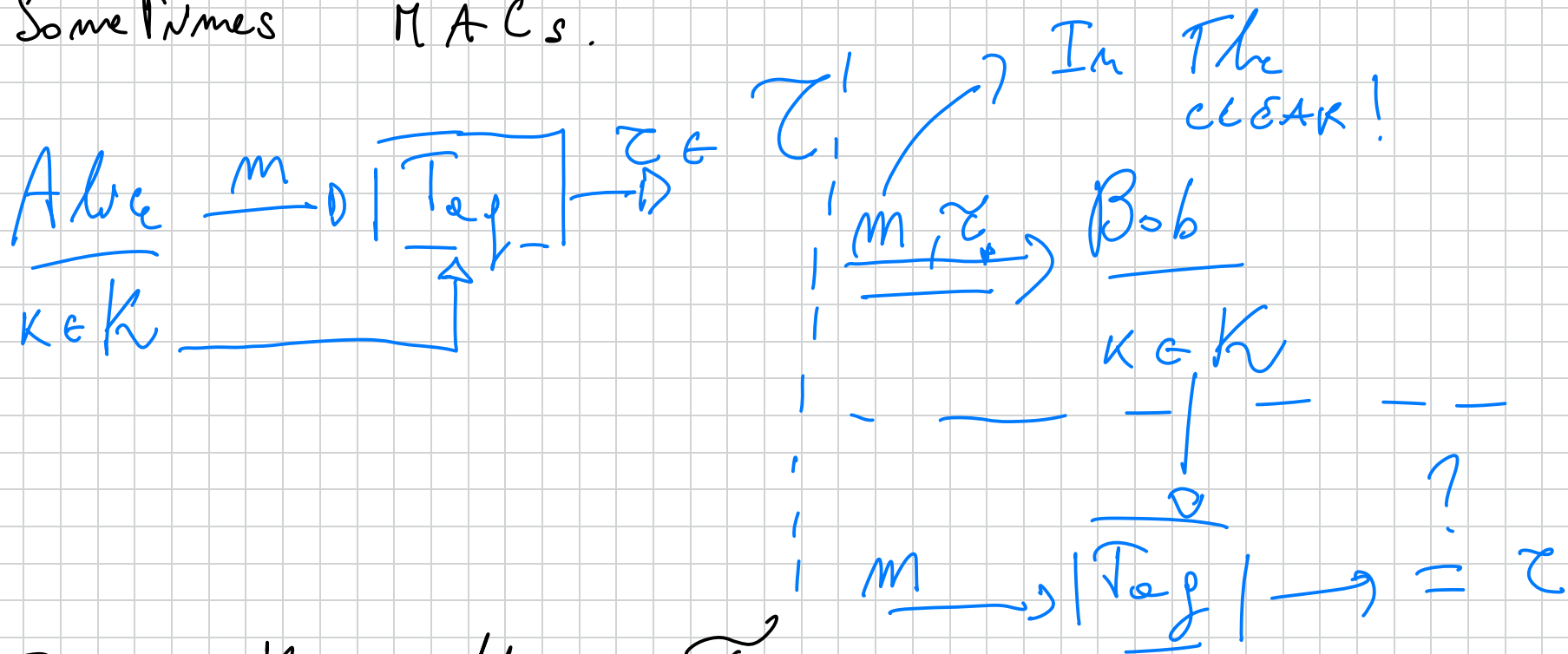
$\equiv$

$$\text{GAME}_A(1, m_0, m_1)$$

This is equivalent to PS.

# MESSAGE AUTHENTICATION CODES

Sometimes MACs.



$\text{Tag} : \mathcal{K} \times \mathcal{M} \rightarrow \mathcal{T}$

$\hookrightarrow$  DETERMINISTIC

SECURITY : ? ? ?

Goal of Eve: Produce some  $m', \tau'$ ,  
without knowing  $k$ , s.t.  $\tau' = \text{Tag}(k, m')$ .

What about: Eve knows a valid pair  $(m, \tau)$   
(i.e.  $\text{Tag}(k, m) = \tau$ ). It should be hard to  
forge  $\tau'$  valid for  $m' \neq m$ .

DEF Let  $\text{Tag}$  be a MAC over  $K, \mathcal{M}, \mathcal{Z}$ . We say  $\text{Tag}$  is

$\epsilon$ -STATISTICALLY secure if  $\forall m, m' \in \mathcal{M}, \forall \tau, \tau' \in \mathcal{Z}$   
 $m \neq m'$   
(ONE-TIME)

$$\Pr[\tau' = \text{Tag}(k, m') \mid \tau = \text{Tag}(k, m)] \leq \epsilon.$$

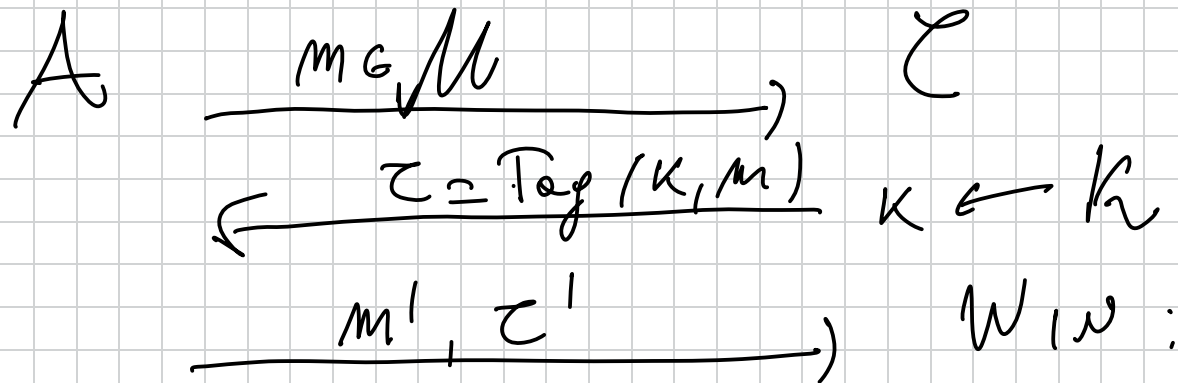
Values of  $\epsilon$ :  $2^{-80}$ ;  $2^{-128}$ .

Exercise.  $\epsilon$ -security is impossible for  $\epsilon = 0$ .

Exercise. One-Time Pad is insecure

$$\text{Tag}(K, m) = z = m \oplus K.$$

Exercise Consider a game as before:



Win:

(i)  $m' \neq m$

(ii)  $\text{Tag}(K, m') = z'$

$\forall A$ :

$$\Pr[\text{win}] \leq \epsilon.$$

We'll construct MACs using some kind of HASH FUNCTIONS.

DEF A family  $H = \{h_K : \mathcal{M} \rightarrow \mathcal{Z} \mid K \in \mathcal{K}\}$  is PAIRWISE INDEPENDENT if  $\forall m, m' \in \mathcal{M}$  Then  $m \neq m'$

$$(h(K, m), h(K, m'))$$

$K$  UNIFORM over  $\mathcal{K}$

is UNIFORM over  $\mathcal{Z}^2 = \mathcal{Z} \times \mathcal{Z}$

This suggests to do:

$$\begin{aligned} \text{Tag}(K, m) &= h(K, m) \\ &= h_K(m) \end{aligned}$$